

ECE 310 Semiconductor Processing Lab 1: Metal Oxide Semiconductor Capacitor

By: Alex Colon

Introduction:

In this lab, our section worked together to get hands-on experience with semiconductor fabrication processes. Specifically, this lab was focused on the fabrication of a Metal Oxide Semiconductor (MOS) capacitor, which consists of a metal gate, oxide insulating layer, and semiconductor substrate. During our time in the clean room, we reviewed standard safety procedures and the inner workings of the machinery used. Our goal was to practice with the machinery, create a MOS capacitor, and test its effectiveness.

MOS capacitors are important because they demonstrate the fundamental electrostatic behavior of semiconductor devices, including charge accumulation, depletion, and inversion, which are critical to the operation of modern integrated circuits. Additionally, understanding MOS capacitors helps us to better understand operations of MOSFETs and their physical principles that allow them to be used in industry.

Methods:

Overview

The fabrication of a MOS capacitor has multiple steps. In this lab, we did AMD cleaning of the silicon, thickness measurement with an ellipsometer, resistivity measurements with a four point probe, dry oxide deposition, plasma dry etching, back side aluminum deposition, front side aluminum contacts, and finally testing (creating C-V plots). This procedure was done over multiple lab sections, and some steps (i.e. cleaning) were repeated.

Day one:

To begin the fabrication process, we started with a standard 2-inch silicon wafer. Before doing any measurements, we needed an AMD clean. The order of operations for this step goes as follows: rinse both sides of the wafer with acetone, rinse again with IPA, rinse off all chemicals with DI water, then dry the wafer off. After the cleaning process, we bake the wafer for 5 minutes (dry bake). This process was used multiple times throughout the lab.

After the wafer is cleaned, it is ready to be measured. Our goal was to measure the initial oxide thickness of the wafer. This was accomplished with an ellipsometer, which uses a laser to analyze the change in polarization of the light reflecting off the wafer. Our desired initial thickness is as close to zero as possible (1-2 nm).

The next measurement was sheet resistivity. The tool we used for this was a four point probe, which gave a measurement for voltage. Additionally, it is known that the current flowing was 100uA and rho (resistivity) has the following relationship:

$$\text{Rho} = (\pi/\ln 2) * (V/I)$$

(With units: $\Omega/\square$)

After getting initial measurements for the wafer, we could start the process of fabricating the capacitor. We began thermal (dry) oxidation using a furnace. A boat carrying the wafers is placed roughly at the halfway point in the furnace, and we supply O2 flow to create oxidation.

Day two:

Day two was started with another AMD clean and dry bake.

Following the clean, we began the process of plasma dry etching. Prior to using the machine (Reactive Ion Etch), it is important to check argon gas pressure, oxygen pressure, nitrogen level, and cooling temp (below 32 degrees C). With the correct set up, we then put the system in standby, and vented the chamber. Controlling pressure is a key part of this, as not doing so could

damage the equipment. Once ready, we placed the wafers faced down in the chamber, then closed and sealed it. This process involves creating a vacuum so the argon, nitrogen, and oxygen create plasma, which is used to etch. The wafers sit in the chamber for five minutes, and then the chamber is depressurized and opened. Once the wafers are removed, the system is returned to a vacuum (this is to preserve the equipment).

After the dry etch, we began our backside aluminum deposition. This was done using the MaSC Nitride Sputter tool. The specifications for this step are: Deposition Rate: 14.5 nm/min. Dep time: 10 min. Pressure: 4E-3 torr. Power: 250W.

Prior to using, we once again needed to check that everything was at its proper operating point. When ready, we begin by turning off the ionization gauge and letting the chamber come to atmosphere (760 torr). We then set the target to be aluminum (for deposition) and set the power as specified above. The wafers were then sealed in and a vacuum was created. The wafer was then transferred to the next chamber, where the deposition process began (following deposition specs). When finished, we once again had to control the pressure and vacuums to prevent damaging the equipment. The power was then restarted to zero, and the throttle was opened.

Day 3:

The next step was aluminum front side deposition. This process used a Cookes evaporator, which evaporates the substance (aluminum) onto the wafer. Before using the machine, we check the utilities and make sure it is ready to operate. The power and pump turned on and the high vac was closed. When ready, we open the chamber and set the mask down followed by the wafer (placed face down). Additionally, a desired amount of aluminum is also placed in the chamber under a shutter.

When ready, the pressure is brought down to 50 mtorr and a vacuum is created. The desired thickness was 150 nm. Bulk air is removed from the chamber and we switch to high vacuum.

When pressure is stabilized, we let sit for five minutes. The gauge is then turned, with a desired pressure of $2e^{-5}$. Current is then ramped up, the shutter is then removed, and the aluminum evaporates onto the wafers.

When done, the current is slowly ramped down. We switch from the pump in order to get out of the high vacuum. Finally, thickness is measured again using the ellipsometer. This was done to calculate the oxide growth between final and initial measurements (taking into account the aluminum growth and adjusting).

Measurements:

Our goal for measurements was to generate CV plots in order to analyze the effectiveness of the MOS capacitors. We placed the wafer in a dark box, and using a micromanipulator we were able to sweep over multiple voltages (-10V to 10V) and generate multiple CV plots.

Results

Day 1 Expected:

Substrate thickness (0.325 mm)

Substrate resistivity (~30 Ohm-cm)

Native oxide thickness (~15 Å)

Oxide thickness - our desired initial thickness was as close to zero as possible (1-2 nm).

The initial thickness we measured was:

Wafer 1: 4.084 +/- 0.727 (nm)

Wafer 2: 4.0237 +/- 0.94 (nm)

Four point probe measurements:

Wafer 1: 11.15 mV

Wafer 2: 11.3 mV

Using the given formula: $\text{Rho} = (\pi/\ln 2) * (V/I)$

Sheet resistivity of wafer 1: 505.36 $\Omega/\square$

Sheet resistivity of wafer 2: 512.16 $\Omega/\square$

Day 2 Expected:

Oxidation thickness: 700 A

Backside aluminum deposition: 150 nm

Thermal oxidation:

The temperature is set at 1100 degrees celsius and we supply O2 flow at 1.584 lpm. The boat is left in the furnace for 20 minutes.

Final measured oxide thickness:

Wafer 1: 86.362 +/- 36.9 (nm)

Wafer 2: 104 +/- 104 (nm)

Backside aluminum deposition:

With a deposition rate of approximately 14.5nm/min, we deposited for approximately 10 minutes.

Day 3 Expected:

Front side aluminum deposition: 150 nm

Measurements:

Low frequency sweep (5kHz)

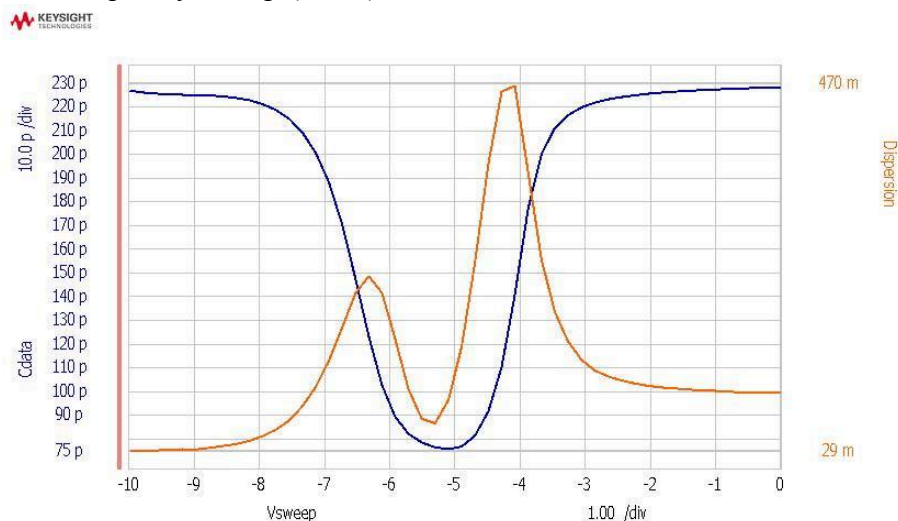


Figure 1: low frequency CV plot

Regions of operation

Accumulation: -10 to -7 V

Depletion: -7 to -4.5 V

Inversion: -4.5 to 0 V

Flat band voltage (point of inflection between accumulation and depletion): -7V

High frequency sweep (500kHz)

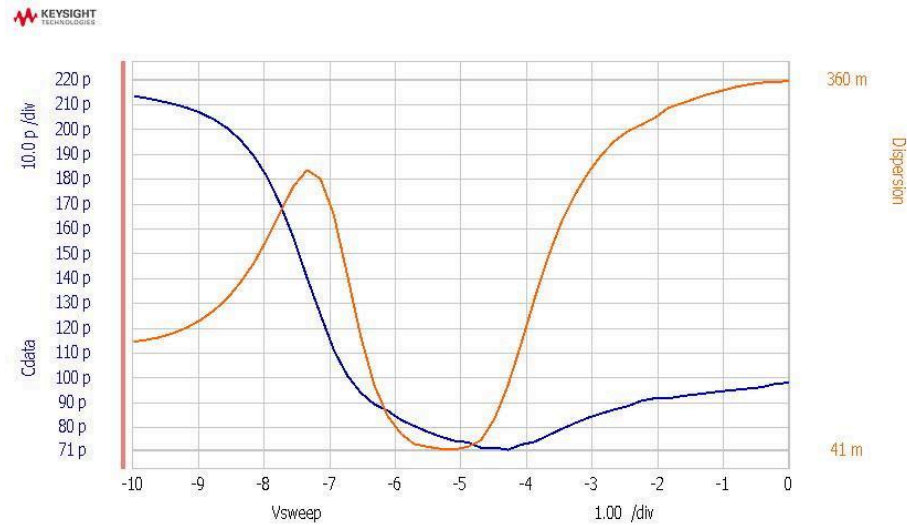


Figure 2: High frequency CV plot

Regions of operation

Accumulation: -10 to -8 V

Depletion: -8 to -4.5 V

Inversion: -4.5 to 0 V

Flat band voltage: -8V

Measured diameter using Optical Microscope:

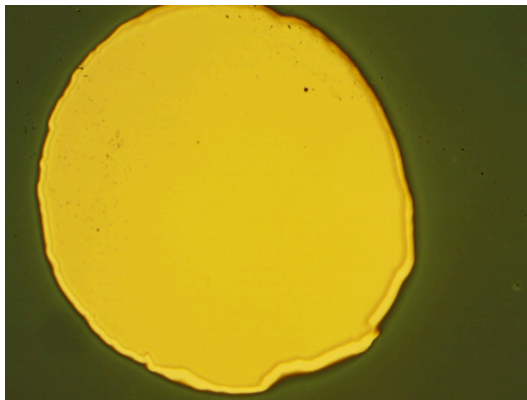


Figure 3: capacitor under optical microscope

Major axis 1.216 mm

Major axis 1.121 mm

Discussion of Results

As seen in our CV plots, we were able to successfully fabricate a working MOS capacitor. The results from each stage of fabrication provide insight into both the quality of our process and the sources of deviation from expected values.

The initial oxide thickness measurements for both wafers (4.084 nm and 4.024 nm) were slightly higher than the expected native oxide thickness of ~ 1.5 nm (15 Å). This small discrepancy is likely due to minor oxidation that occurred during handling and transport of the wafers prior to measurement, which is common in semiconductor fabrication. The AMD cleaning process was intended to minimize this, but trace oxidation is difficult to eliminate entirely.

The sheet resistivity values calculated from the four-point probe were 505.36 and 512.16 $\Omega/\square$ for wafers 1 and 2, respectively. Converting to bulk resistivity using the known substrate thickness of 0.325 mm yields values reasonably close to the expected $\sim 30 \Omega\cdot\text{cm}$, confirming the wafers were within acceptable doping specifications.

The target oxide thickness after dry oxidation was 700 Å (70 nm). Wafer 1 measured 86.36 nm and Wafer 2 measured 104 nm, both exceeding the target. The large uncertainty on Wafer 2 (± 104 nm) is notable and suggests measurement inconsistency, possibly due to non-uniform oxide growth, ellipsometer calibration error, or surface contamination. The overgrowth on both wafers may be attributed to slight variations in furnace temperature uniformity or O₂ flow rate during the 20 minute oxidation window. A thicker oxide than intended will shift the CV curve and affect the flat-band voltage, which may help explain the observed flat-band voltages of -7V and -8V in the low and high frequency sweeps, respectively.

Figure 1 (low frequency, 5 kHz) clearly shows all three regions of MOS capacitor operation. In accumulation (-10 to -7 V), the negative gate voltage attracts majority carriers (holes, indicating a p-type substrate) to the oxide interface, resulting in high capacitance near the oxide capacitance. In the depletion region (-7 to -4.5 V), majority carriers are repelled and a depletion layer forms, causing capacitance to decrease. In the inversion region (-4.5 to 0 V), the low-frequency plot shows capacitance recovering toward oxide capacitance, which is the expected behavior at low frequencies: minority carriers have sufficient time to respond to the AC signal and contribute to the capacitance.

Figure 2 (high frequency, 500 kHz) mirrors the low-frequency plot through accumulation and depletion, but diverges in the inversion region, where capacitance remains low and plateaus. This is the hallmark of high frequency CV behavior: minority carriers cannot generate and recombine fast enough to follow the rapid AC signal, so they do not contribute to the measured capacitance. This distinction between the two plots is a strong indicator that our device is functioning as expected.

The flat-band voltage was measured at -7V (low frequency) and -8V (high frequency). A negative flat-band voltage shift from the ideal 0V suggests the presence of positive fixed oxide charges at the Si-SiO₂ interface, which is a common outcome of dry oxidation processes. These charges attract additional negative charge in the semiconductor, requiring a more negative applied voltage to achieve the flat-band condition.

The optical microscope measurements showed the capacitor contact to be slightly elliptical, with major axes of 1.216 mm and 1.121 mm, giving an effective contact area (using parallel plate

capacitor formula) of $A = \pi \times (0.608 \text{ mm}) \times (0.5605 \text{ mm}) = 1.070 \text{ mm}^2 = 1.070 \times 10^{-6} \text{ m}^2$. Using this area alongside the accumulation capacitance values read from the CV plots ($\sim 225 \text{ pF}$ at low frequency and $\sim 210 \text{ pF}$ at high frequency) and the relation $t = (\epsilon_0 \times \epsilon_r \times A) / C_{\text{ox}}$, with $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$ and $\epsilon_r = 3.9$ for SiO_2 , the extracted oxide thicknesses are $t = 16.4 \text{ nm}$ (low frequency) and $t = 17.6 \text{ nm}$ (high frequency). These values are notably lower than the ellipsometer measurements of 86.4 nm and 104 nm from Day 2. This large discrepancy is likely due to the aluminum contacts not covering the full elliptical area assumed in the calculation, if the effective contact area is smaller than measured optically, C_{ox} would appear larger, pulling t down significantly. Non-ideal contact adhesion, edge effects during aluminum deposition, or errors in reading the accumulation plateau from the CV curve could also contribute to this inconsistency.

Conclusion

This lab served as an introduction to semiconductor processing, teaching both technical uses of machinery and procedure for fabricating a MOS capacitor. We started with a silicon wafer and proceeded to clean, measure, deposit oxide (dry thermal oxidation), dry etch with plasma using an RIE, deposit back side aluminum using the MaSC Nitride Sputter tool, deposit front side aluminum with a mask using a Cooke's evaporator, and finally we created CV plots and took measurements of our capacitors. Each step we learned how the machinery operates on a physical level and how to use the machinery for fabrication. In the end we successfully fabricated MOS capacitors and verified that they work.